Surya Sai Satya Likhith Vardhineedi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.04
Sri Chaitanya Junior Kalasala GUDAVALLI	Class XII (Board of Intermediate Education Andhra Pradesh)	2021	93.80%
DR K K R GOWTHAM H S GUDAVALLI	Class X (Board of Secondary Education Andhra Pradesh)	2019	9.30%

INTERNSHIP

Rashtriya Ispat Nigam Limited (RINL), Visakhapatnam Steel Plant

ML Project Trainee

- **Developed a machine learning-based prediction model** for blast furnace skin temperature using process parameters such as cold blast flow, pressure, top temperature, gas composition, and oxygen flow.
- **Performed exploratory data analysis and feature engineering** to identify relationships between furnace operating conditions and temperature behavior, supporting data-driven process monitoring and optimization.
- Built and evaluated a **Random Forest Regression** model, achieving an **R^2 score of 0.809** on the reported test results.

THESIS

Uncertainty Quantification in Autonomous Navigation

- **Investigating uncertainty-aware perception for autonomous navigation**, focusing on uncertainty estimation in modern object detection and segmentation models used for autonomous systems.
- **Studying and evaluating uncertainty quantification methods**, including Bayesian and approximate Bayesian approaches, for models such as **RT-DETR, DINO, and other transformer-based vision architectures**.
- **Developing an experimental framework** to quantify and analyze **aleatoric and epistemic uncertainty**, evaluating how uncertainty estimates relate to object-detection reliability, prediction confidence, and potential safety-critical scenarios in autonomous navigation

PROJECTS

EV Battery Health Analysis and Predictive Maintenance

- **Engineered 16 physics-informed degradation features** from 2,794 lithium-ion discharge cycles, incorporating voltage, current, temperature, energy, impedance, discharge duration, and rolling statistics.
- **Developed and evaluated Ridge Regression, Random Forest, and XGBoost models** for State of Health (SoH) and Remaining Useful Life (RUL) estimation using time-aware 70/30 validation.
- **Designed a predictive maintenance decision framework** that converts battery health predictions into **HEALTHY, MONITOR, and REPLACE** states; XGBoost achieved **R^2 = 0.9821 with 1.8% relative SoH error**.

Fraud Detection & Risk Scoring

- **Engineered a leakage-safe fraud detection pipeline** for **60,000 transaction records**, using EDA, SQL queries, and statistical tests to identify fraud patterns.
- **Developed and compared Logistic Regression, Random Forest, and XGBoost models** for fraud classification using chronological train/validation/test splits and imbalance-aware metrics; XGBoost achieved **F1 = 0.453, Precision = 0.349**, while Logistic Regression achieved **Recall = 0.904 and ROC-AUC = 0.966**.
- **Implemented SHAP-based explainability and risk scoring**, converting fraud probabilities into Low/Medium/High risk tiers, and deployed transaction-level prediction pipeline using **FastAPI and Streamlit**.

Credit Risk & Loan Default Prediction

- **Engineered a credit-risk pipeline** for 20,000 synthetic Indian loan records using median imputation, winsorisation, feature scaling, and domain-based DTI, Delinquency Score, EMI-to-Income, and Income-per-Dependent features.
- **Developed and evaluated Logistic Regression, Random Forest, XGBoost, and SMOTE-based classification models**, using ROC-AUC, Precision-Recall metrics, class-weighting/SMOTE, and validation-based decision-threshold optimization; the best test ROC-AUC was **0.6123 with 75% default recall**.
- **Implemented SHAP-based model explainability and deployed an interactive Streamlit dashboard** for borrower-level default probability, risk classification, model comparison, and feature contribution analysis.

SKILLS

Languages: Python, SQL, C/C++

Data Visualization: Matplotlib, Seaborn, Power BI, Tableau

Data Science: NumPy, Pandas, Scipy, EDA, Statistics, Probability, Statistical Modelling, A/B Testing

Machine Learning: Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, OpenCV, Hugging Face

ML Algorithms: Linear/Logistic Regression, Decision Trees, Random Forest, KNN, SVM, Gradient Boosting, PCA, Clustering

Databases & Tools: MySQL, Git/GitHub, Jupyter, VS Code, Google Colab, Advanced Excel

GenAI & LLMs: Prompt Engineering, RAG, AI Agents, Agentic Workflows

RELEVANT COURSES

Foundations of Machine Learning, Advanced Computer Networks, Autonomous Navigation, Data Science Analysis, Deep Learning